

Class Activity Queue

temp = 15

1. Show the output of the following code: (5 points)

20, 42, 5, 16, 30

```
queueType<int> queue;
int temp = 20;

queue.addQueue(15);
queue.addQueue(8);
queue.addQueue(temp);
queue.addQueue(42);
temp = queue.front();
queue.deleteQueue();
queue.addQueue(temp - 10);
queue.addQueue(16);
queue.addQueue(2 * temp);
queue.deleteQueue();

cout << "Queue elements: ";
while (!queue.isEmptyQueue())
{
    cout << queue.front() << " ";
    queue.deleteQueue();
}
cout << endl;
cout << "temp = " << temp << endl;
```

OUTPUT:

Queue elements: 20, 42, 5, 16, 30
temp = 15

2. Show the answer to the following questions: (5 points)

Suppose that **queue** is a **queueType** object and the size of the array implementing **queue** is 65. Also, suppose that the value of **queueFront** is 35 and the value of **queueRear** is 60. (9)

- a. How many elements are in **queue**? 25 elements in the queue
- b. What are the values of **queueFront** and **queueRear** after adding an element to **queue**? **queueFront** is still 35, and **queueRear** is now 61
- c. What are the values of **queueFront** and **queueRear** after removing an element from **queue**? original **queue**

queueFront is now 36 and **queueRear** is still 60